External Plural Effect

This notebook investigates the external plural effect (ExtPE) in German compounding as defined by Schäfer & Pankratz 2018:

> A plural on the entire compound (formally on the head constituent) might trigger the use of a pluralic linking element. We call this the external plural effect.

Background

Schäfer & Pankratz 2018 assembled a German compound dataset of N=9415 compounds built from 48 first constituent (N1s), up to ten N1 for each of the seven pluralic linkers *-n-*, *-en-*, *-e-*, *-"e-*, *-"-*, *-er-*, *-"er-*. All N1s are countable nouns which are productively used in compounds both with and without the pluralic linking element. For each N1, up to 100 compound **tokens** with a non-pluralic (mostly zero or *-s-*) and up to 100 compound **tokens** with the pluralic linker were retrieved from DECOW16A. Since tokens and not types were counted, there is a number of duplicated constituent combinations (CC) in the data, though sometimes with alternating linkers. Besides, the original wordforms were preserved, and so compounds both in their Sg and Pl form appear, sometimes within the duplicated CC. Thus for example, the CC 'Apfel' + 'Baum' features occurrences 'Apfelbaum', 'Apfelbäume', 'Äpfelbaum', and 'Äpfelbäume' in the data. Note that no frequency threshold on the compounds was set while sampling the data.

What Schäfer & Pankratz 2018 were interested in regarding the ExtPE was whether the formal Pl form of a compound increases the probability of the pluralic linker element to occur. Going back to the example above, the question was: if the Sg form of 'Apfel' + 'Baum' is 'Apfelbaum', will the Pl form statistically tend to prefer 'Äpfelbäume' over 'Apfelbäume'?

To investigate this effect, Schäfer & Pankratz 2018 utilized the following methodology.

1. First, they created a 2x2 contingency table over the data with the following axes: non-pluralic vs. pluralic linker, and Sg vs. Pl compound form. The resulting table is below:

	Pl	Sg

non-pluralic linker	1574	3133
pluralic linker	1474	3234

2. Second, they applied the Chi2 test to identify whether there is a relation between the two variables (the "plurality" of the linker vs. the formal number of the compound). The test produced a significant result at sig = 0.05 $(χ2 = 4.786, df = 1, p = 0.029)$.
3. Then, they measured the effect size with Cramer's V and revealed that it is quite low at $v = 0.023$.
4. Finally, the inspection of the symmetrical (standardised) residuals revealed that the diverging data points followed the wrong direction, that is, there were more non-pluralic linkers in Pl compounds that expected by chance (-2.210).

Schäfer & Pankratz 2018 conclude: even though the p-value is (still) significant, the Cramer's V and the residual values reveal that even if there is some weak association, it is opposite from expected: compounds more often have non-pluralic linkers in the Sg form.

As a confirmation, the preceding Pankratz 2017 thesis was based on the same methodology and gave similar results for the ExtPE on a different dataset. Pankratz found only four individual N1s that undergo the ExtPE in the desired direction, and four that went in the opposite direction. The ExtPE hypothesis was still refuted because no effect was detected for the entire dataset.

Objective

When scrutinizing the dataset from Schäfer & Pankratz 2018, a few groups of compounds could be detected that could erode the overall effect over the entire dataset:

- Compounds that only appear in one number. Under no observed number variation, we cannot make judgments about the behavior of these compounds in the opposite number.
- Compounds that (potentially) undergo the internal plural effect (IntPE). Since the IntPE works on a lexical level, it should be present in both numbers, so that both in Sg and Pl forms the pluralic linker is preferred (examples from the dataset: 'Äpfelkiste' vs. 'Äpfelkisten', 'Liederabend' vs. 'Liederabende'). Thus, the IntPE may interfere with the external plural effect and make the latter undetectable.
- Compounds whose N1s may take on the pluralic linker due to other constraints (examples from the dataset: 'Katzenfell' vs. 'Katzenfelle' because

of schwa, 'Kinderbuch' vs. 'Kinderbücher' as person designation). Again, if the given N1 mostly takes a pluralic linker in its Sg form, it will most likely do so in Pl, making the ExtPE undetectable.

Hypothesis 1 was the following: *after removing these "problematic" groups from the data, the ExtPE should be observable in the remaining compounds*.

Hypothesis 2 focuses on the effect of the "problematic" groups on ExtPE: *gradually letting the problematic groups back in should weaken the ExtPE down to its vanishing*.

On the other hand, the ExtPE may not be present the standard German, since variation of type 'Apfelbaum' -- 'Äpfelbäume' is ungrammatical in standard German. If the ExtPE exists, it should be an immediate product of colloquial language, maybe even unintentional. This is indirectly supported by the fact that the compounds in the data did not undergo any frequency filtering. Therefore, we proposed the **Hypothesis 3**: *most (if not all) manifestations of the ExtPE will vanish after frequency filtering. A more severe filtering will lead to a more pronounced absence of the effect*.

Data

Convention: *(N=x)* refers to the number of compound **tokens** in the dataset, *(UniqN=x)* refers to the number of unique compound **lemmata** in the dataset, *(UniqCN=x)* refers to the number of unique constituent combinations in the dataset (regardless of the linker), *(UniqN1N=x)* refers to the number of unique N1 **lemmata** in the dataset, "compounds" refers to "compound tokens" unless specified otherwise.

We prepared three modifications of the original Schäfer & Pankratz 2018 dataset resulting after removing the problematic groups gradually.

- `ic_alt` (**I**dentical **C**onstituent combinations, **ALT**ernating): compounds built from non-unique constituent combinations (regardless of the linker) appearing both in the Sg and the Pl forms | $(N=2318, UniqN=1246, UniqCN=869)$.
- `ic_alt_nipe` (**I**dentical **C**onstituent combinations, **ALT**ernating, **N**o **I**nt**PE**): compounds built from non-unique constituent combinations (regardless of the linker) appearing both in the Sg and the Pl forms and not undergoing the IntPE | $(N=1931, UniqN=1043, UniqCN=724)$.
- `ic_alt_nipe_npll` (**I**dentical **C**onstituent combinations, **ALT**ernating, **N**o **I**nt**PE**, **N**o **PL**uralic **L**inker allowed): compounds built from non-unique constituent combinations (regardless of the linker) appearing both in the Sg and the Pl forms and not undergoing the IntPE whose N1 may not attach a pluralic linker | $(N=1236, UniqN=665, UniqCN=468, UniqN1N=30)$.

In parallel, we defined three levels of frequency (the word counts (WCs) were obtained from DeReKo following the procedure outlined in Shmalts 2026, accessed mid June 2026):

- `lfreq` (low frequency): word count (WC) > 0, as in the original data.
- `mfreq` (moderate frequency): WC $>= 20$.
- `hfreq` (high frequency): WC $>= 100$.

During the studies, each of the three datasets was dynamically filtered to conform to the three frequency levels. Thus, nine unique data samples appear in the studies.

Workflow

To evaluate the proposed hypothesis, we run four studies.

Study 1

Study 1 replicates the original ExtPE study from Schäfer & Pankratz 2018 on the nine possible data samples (three datasets x three frequency levels). It answers the general question whether the ExtPE is present (in any of the samples). Comparing the individual results from the samples will additionally give an overall idea of the impact of the problematic groups and the frequency level on ExtPE's presence and strengths.

Study 1 contributes to **Hypothesis 1**. Expected outcome: ExtPE will be confirmed on less "contaminated" datasets in the lower frequency setting.

It also indirectly contributes to **Hypothesis 2** and **Hypothesis 3**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE.

Study 2

Study 2 expands study 1 and investigates the ExtPE for each individual N1 in each of the nine data samples, following Pankratz 2017. It aims at answering whether certain N1s are more sensitive to the ExtPE than the others.

Study 2 contributes to **Hypothesis 1**. Expected outcome: ExtPE will be confirmed on less "contaminated" datasets in the lower frequency setting with some N1s.

It also indirectly contributes to **Hypothesis 2** and **Hypothesis 3**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE with some or all N1s.

Study 3

Study 3 focuses on the distribution of non-pluralic and pluralic linker in compounds in the Sg form only. It contrasts the distribution in the "cleanest" `ic_alt_nipe_npll` vs the two other datasets with an increasing level of "contamination" by the problematic groups.

Study 3 contributes to **Hypothesis 2** and provides a more detailed account of the effect of the problematic groups on ExtPE. Expected outcome: more "contaminated" datasets will have a larger share of pluralic linkers in the Sg form (due to more possible sources).

Study 4

Study 4 focuses on the distribution of non-pluralic and pluralic linker in compounds in the Pl form only. It contrasts the distribution in the low frequency setting vs the moderate/high frequency setting, within each of the three dataset.

Study 4 contributes to **Hypothesis 3** and provides a more detailed account of the effect of the frequency setting on ExtPE. Expected outcome: the share of the pluralic linkers in compounds in the Pl form will be larger in in the low frequency setting (since most of the ExtPE manifestations are a product of immediate language).

Results

Summary

The findings from the four studies **refute Hypothesis 1** that the ExtPE generally exists in less "contaminated" data. Conversely, they confirm **Hypothesis 2 and Hypothesis 3** that any manifestations of the ExtPE are a product of the low frequency setting, better detectable in less "contaminated data".

Note that the Cramer's V value did not cross the value of *0.1* in any of the tests. That means that, even if an effect is statistically significant, the effect size is very small.

Study 1

> Summary: Study 1 provides direct evidence **against Hypothesis 1** and indirect evidence **for Hypothesis 2** and **for Hypothesis 3**.

Study 1 contributes to **Hypothesis 1**. Expected outcome: ExtPE will be confirmed on less "contaminated" datasets in the lower frequency setting.

Actual outcome: The ExtPE does not reach a statistical significance in any of the

nine data samples. The p-values and the Cramer's V values are comparable across the three datasets along all three frequency settings.

It also indirectly contributes to **Hypothesis 2** and **Hypothesis 3**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE.

Actual outcome: the p-values increase, while the Cramer's V values decrease consistently within each dataset, as the frequency level of the dataset is increased. This points out that ExtPE is much weaker in moderate/high frequency settings, which indirectly points towards the expected frequency effect on the ExtPE.

It also partly contributes to **Hypothesis 2**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE.

Actual outcome: the p-values increase, while the Cramer's V values decrease consistently within each dataset, as the frequency level of the dataset is increased. This points out that ExtPE is much weaker in moderate/high frequency settings, which indirectly points towards the expected frequency effect on the ExtPE.

Study 2

> Summary: Study 2 provides partial direct evidence **for Hypothesis 1** and indirect evidence **for Hypothesis 2** and **for Hypothesis 3**.

Study 2 contributes to **Hypothesis 1**. Expected outcome: ExtPE will be confirmed on less "contaminated" datasets in the lower frequency setting with some N1s.

Actual outcome: N1s 'Apfel' and 'Bucht' undergo the ExtPE in the desired direction at marginal significance level sig = 0.1 (increased p-value threshold to account for the small sample sizes), but only in `ic_alt_nipe_npll` and `ic_alt_nipe` and only in the low frequency setting ('Apfel': $χ2 = 3.242, df = 1, p = 0.072, v = 0.265$; 'Bucht': $χ2 = 2.716, df = 1, p = 0.099, v = 0.243$). N1 'Person' undergoes the ExtPE in the desired direction at significance level close to sig = 0.05 in all three datasets, but, again, only in the low frequency setting ($χ2 = 3.967, df = 1, p = 0.046, v = 0.245$).

It also indirectly contributes to **Hypothesis 2** and **Hypothesis 3**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE with some or all N1s.

Actual outcome: the fact that the marginal significance disappears with 'Apfel' and 'Bucht' in the most "contaminated" dataset `ic_alt` indirectly points towards the

effect of the problematic groups on the ExtPE. The fact that the (marginal) significance disappears with all three N1s in the moderate/high frequency settings indirectly points towards the effect of frequency on the ExtPE.

The significant results of N1s 'Apfel' and 'Person' confirm finding of Pankratz 2017. However, none of the other six "significant" N1s reported by Pankratz 2017 reach statistical significance in any of the nine data samples.

Study 3

> Summary: Study 3 provides direct evidence **for Hypothesis 2**.

Study 3 contributes to **Hypothesis 2**. Expected outcome: more "contaminated" datasets will have a larger share of pluralic linkers in the Sg form (due to more possible sources).

Actual outcome: while comparison of the linker distributions over the compounds in the Sg form in `ic_alt_nipe_npll` vs `ic_alt_nipe` did not produce statistical significance, comparing the "cleanest" `ic_alt_nipe_npll` and the most "contaminated" `ic_alt` reveals a statistically significant effect at sig = 0.005 ($χ2 = 10.394, df = 1, p = 0.001, v = 0.076$). The examination of the symmetrical (standardised) residuals reveals that the diverging data points followed the expected direction, that is, compounds in the Sg form in `ic_alt` have a larger share of pluralic linkers, and those in `ic_alt_nipe_npll` a smaller share of pluralic linkers, than expected by chance (3.274). This means that compounds in `ic_alt` have more sources for the pluralic linker to appear in the Sg form. The finding supports the indirect evidence for Hypothesis 2 from Study 2, and confirms the effect.

Study 4

> Summary: Study 4 provides direct evidence **for Hypothesis 3** and indirect evidence **for Hypothesis 2**.

Study 4 contributes to **Hypothesis 4**. Expected outcome: the share of the pluralic linkers in compounds in the Pl form will be larger in in the low frequency setting (since most of the ExtPE manifestations are a product of immediate language).

Actual outcome: comparison of the linker distributions over the compounds in the Pl form in difference frequency settings produced statistical significance only in the `ic_alt_nipe_npll` dataset: low vs. moderate frequency reaches significance at sig = 0.05, and low vs. high frequency at sig = 0.005 (low vs.

moderate: $χ2 = 4.376, df = 1, p = 0.036, v = 0.066$; low vs. high: $χ2 = 7.755, df = 1, p = 0.005, v = 0.09$). The examination of the symmetrical (standardised) residuals reveals that the diverging data points followed the expected direction, that is, compounds in the Pl form inserted pluralic linkers more frequently in the low frequency setting and less frequently in the moderate/high than expected by chance (2.158 and 2.854, respectively). This finding supports the indirect evidence for Hypothesis 3 from Studies 1 and 2, and confirms the effect.

The fact that the more "contaminated" datasets did not exhibit a significant difference additionally speaks for Hypothesis 2 and may be explained by that the additional sources of pluralic linkers there erode the effect. This is attested by the fact that, while insignificant, the p-values still decrease, and the Cramer's V values increase there consistently with increasing the frequency level.

Conclusion

While the theoritized ExtPE is not strong or consistent enough to be acknowledged as an established phenomenon in German, there exist arbitrary and scattered weak manifestations thereof. These manifestations occur more frequently in the low frequency setting that in the moderate/high frequency setting, with a statistically significant difference. They are found prevalently in compounds that would otherwise not insert a pluralic linkers, with a statistically significant difference as compared to compounds that may do so. We conclude: while inconsistently and pronounced weakly, certain sparks of the ExtPE may occur in immediate colloquial language. If it happens, the object of the ExtPE is more likely a compounds that would otherwise not insert a pluralic linker. In compounds that may do so, the ExtPE is masked and not detectable.

Study 1

Study 1 replicates the original ExtPE study from Schäfer & Pankratz 2018 on the nine possible data samples (three datasets x three frequency levels). It answers the general question whether the ExtPE is present (in any of the samples). Comparing the individual results from the samples will additionally give an overall idea of the impact of the problematic groups and the frequency level on ExtPE's presence and strengths.

Study 1 contributes to **Hypothesis 1**. Expected outcome: ExtPE will be confirmed on less "contaminated" datasets in the lower frequency setting.

It also indirectly contributes to **Hypothesis 2** and **Hypothesis 3**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE.

The following table provides an overview over the results of Study 1. For each combination of the dataset and the frequency setting, the results of the Chi2 test over the resulting sample is given. We can see that none of the p-values are even near low.

```
# 3D res DF: dataset x freq x chi2 test metrics
chi2_summary = pd.concat(chi2_summaries, keys=keys)  \
    .swaplevel(0, 1)    \
    .sort_index(level=0, sort_remaining=False)  \
    .reindex(chi2df.index, level=0)
savetsv(chi2_summary, st1dir, "chi2_summary")
chi2_summary
```

		lfreq	mfreq	hfreq
N	ic_alt_nipe_npll	1235.000	808.000	694.000
	ic_alt_nipe	1930.000	1248.000	1076.000
	ic_alt	2317.000	1516.000	1308.000
df	ic_alt_nipe_npll	1.000	1.000	1.000
	ic_alt_nipe	1.000	1.000	1.000
	ic_alt	1.000	1.000	1.000
chi2	ic_alt_nipe_npll	1.785	0.049	0.026
	ic_alt_nipe	2.051	0.139	0.000
	ic_alt	1.884	0.175	0.012
p	ic_alt_nipe_npll	0.181	0.824	0.872
	ic_alt_nipe	0.152	0.710	1.000
	ic_alt	0.170	0.676	0.914
cramer_v	ic_alt_nipe_npll	0.038	0.008	0.006
	ic_alt_nipe	0.033	0.011	0.000
	ic_alt	0.029	0.011	0.003
resids_cong	ic_alt_nipe_npll	1.394	0.296	-0.242
	ic_alt_nipe	1.478	0.430	0.028
	ic_alt	1.414	0.470	0.163
resids_incong	ic_alt_nipe_npll	-1.394	-0.296	0.242
	ic_alt_nipe	-1.478	-0.430	-0.028
	ic_alt	-1.414	-0.470	-0.163

We can also observe that there is no noticeable difference between the p-values across the datasets (along the same frequency levels), nor any trend. However, there is a clear trend toward increasing p-value with the increasing frequency level, for all three datasets. Similarly, the Cramer's V values drop consistently as the frequency level is increased.

```
chi2_summary.loc[["p", "cramer_v"]]
```

		lfreq	mfreq	hfreq
p	ic_alt_nipe_npll	0.181	0.824	0.872
	ic_alt_nipe	0.152	0.710	1.000
	ic_alt	0.170	0.676	0.914
cramer_v	ic_alt_nipe_npll	0.038	0.008	0.006
	ic_alt_nipe	0.033	0.011	0.000
	ic_alt	0.029	0.011	0.003

Study 1 Results

> Summary: Study 1 provides direct evidence **against Hypothesis 1** and indirect evidence **for Hypothesis 2** and **for Hypothesis 3**.

Study 1 contributes to **Hypothesis 1**. Expected outcome: ExtPE will be confirmed on less "contaminated" datasets in the lower frequency setting.

Actual outcome: The ExtPE does not reach a statistical significance in any of the nine data samples. The p-values and the Cramer's V values are comparable across the three datasets along all three frequency settings.

It also indirectly contributes to **Hypothesis 2** and **Hypothesis 3**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE.

Actual outcome: the p-values increase, while the Cramer's V values decrease consistently within each dataset, as the frequency level of the dataset is increased. This points out that ExtPE is much weaker in moderate/high frequency settings, which indirectly points towards the expected frequency effect on the ExtPE.

Study 2

Study 2 expands study 1 and investigates the ExtPE for each individual N1 in each of the nine data samples, following Pankratz 2017. It aims at answering whether certain N1s are more sensitive to the ExtPE than the others.

Study 2 contributes to **Hypothesis 1**. Expected outcome: ExtPE will be confirmed on less "contaminated" datasets in the lower frequency setting with some N1s.

It also indirectly contributes to **Hypothesis 2** and **Hypothesis 3**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE with some or all N1s.

```
n1s = sp18_sample["n1"].unique()
n1s
```

```
<StringArray>
[      'apfel',         'bad',        'ball',        'bett',
'bild',
       'brett',       'brief',        'buch',       'bucht',
'dämon',
          'ei', 'eigenschaft',     'element',        'frau',
'gerät',
    'geräusch',    'geschenk',     'gitarre',        'hand',
'haus',
        'hemd',        'hund',       'katze',        'kind',
'kommune',
        'lied',        'loch',      'mutter',       'nagel',
'nation',
         'ohr',      'person',     'produkt',   'projektor',
'rad',
     'schloss',     'schwert',   'schwester',       'sonne',
'stadt',
       'vater',       'vogel',         'weg',        'wurm',
'zahn',
       'zitat']
Length: 46, dtype: str
```

Only three N1s reached a marginal statistical significance at sig = 0.1 (increased p-value threshold to account for the small sample sizes).

Below is a table with the results of the Chi2 test for N1s that achieved a marginal statistical significance at sig = 0.1 (increased p-value threshold to account for the smaller sample sizes).

```
chi2_summary = pd.concat(chi2_summaries_conc, axis=1)
savetsv(chi2_summary, st2dir, "chi2_summary")
chi2_summary
```

		person lfreq	bucht lfreq	apfel lfreq	mfreq	hfreq
N	ic_alt_nipe_npll	51.000	46.000	46.000	33.000	26.000
	ic_alt_nipe	51.000	46.000	46.000	33.000	26.000
	ic_alt	66.000	49.000	52.000	33.000	26.000
df	ic_alt_nipe_npll	1.000	1.000	1.000	1.000	1.000
	ic_alt_nipe	1.000	1.000	1.000	1.000	1.000
	ic_alt	1.000	1.000	1.000	1.000	1.000
chi2	ic_alt_nipe_npll	3.780	2.716	3.242	0.136	1.187
	ic_alt_nipe	3.780	2.716	3.242	0.136	1.187
	ic_alt	3.967	2.366	2.233	0.136	1.187
p	ic_alt_nipe_npll	0.052	0.099	0.072	0.712	0.276
	ic_alt_nipe	0.052	0.099	0.072	0.712	0.276
	ic_alt	0.046	0.124	0.135	0.712	0.276
cramer_v	ic_alt_nipe_npll	0.272	0.243	0.265	0.064	0.214
	ic_alt_nipe	0.272	0.243	0.265	0.064	0.214
	ic_alt	0.245	0.220	0.207	0.064	0.214
resids_cong	ic_alt_nipe_npll	2.228	1.951	2.107	0.821	1.323
	ic_alt_nipe	2.228	1.951	2.107	0.821	1.323
	ic_alt	2.246	1.836	1.776	0.821	1.323
resids_incong	ic_alt_nipe_npll	-2.228	-1.951	-2.107	-0.821	-1.323
	ic_alt_nipe	-2.228	-1.951	-2.107	-0.821	-1.323
	ic_alt	-2.246	-1.836	-1.776	-0.821	-1.323

'Apfel' and 'Bucht' reach marginal significance only in the `ic_alt_nipe_npll` and `ic_alt_nipe` datasets, and only in the low frequency setting.

```
chi2_summary.loc[["p", "cramer_v"]][["apfel", "bucht"]]
```

		apfel			bucht
		lfreq	mfreq	hfreq	lfreq
p	ic_alt_nipe_npll	0.072	0.712	0.276	0.099
	ic_alt_nipe	0.072	0.712	0.276	0.099
	ic_alt	0.135	0.712	0.276	0.124
cramer_v	ic_alt_nipe_npll	0.265	0.064	0.214	0.243
	ic_alt_nipe	0.265	0.064	0.214	0.243
	ic_alt	0.207	0.064	0.214	0.220

'Person' reaches (marginal) significance in all three datasets, but also only in the low frequency setting.

```
chi2_summary.loc[["p", "cramer_v"]]["person"]
```

		lfreq
p	ic_alt_nipe_npll	0.052
	ic_alt_nipe	0.052
	ic_alt	0.046
cramer_v	ic_alt_nipe_npll	0.272
	ic_alt_nipe	0.272
	ic_alt	0.245

The reason N1s 'Person' and 'Bucht' have only an `lfreq` column is because after frequency filtering, the remaining compounds with these N1s ceased to alternate between non-pluralic and pluralic linker in different numbers, always inserting the latter. That led to corrupted contingency tables, so no result could be obtained for these N1s in the moderate/high frequency setting. It is, however, clear that no alternation means no ExtPE whatsoever.

```
sign_n1s_mfreq["is_linker_pl"].unique()
```

```
array([ True])
```

Study 2 Results

> Summary: Study 2 provides partial direct evidence **for Hypothesis 1** and indirect evidence **for Hypothesis 2** and **for Hypothesis 3**.

Study 2 contributes to **Hypothesis 1**. Expected outcome: ExtPE will be confirmed on less "contaminated" datasets in the lower frequency setting with some N1s.

Actual outcome: N1s 'Apfel' and 'Bucht' undergo the ExtPE in the desired direction at marginal significance level sig = 0.1 (increased p-value threshold to account for the small sample sizes), but only in `ic_alt_nipe_npll` and `ic_alt_nipe` and only in the low frequency setting ('Apfel': $χ2 = 3.242, df = 1, p = 0.072, v = 0.265$$χ2 = 3.242, df = 1, p = 0.072, v = 0.265$; 'Bucht': $χ2 = 2.716, df = 1, p = 0.099, v = 0.243$$χ2 = 2.716, df = 1, p = 0.099, v = 0.243$). N1 'Person' undergoes the ExtPE in the desired direction at significance level close to sig = 0.05 in all three datasets, but, again, only in the low frequency setting ($χ2 = 3.967, df = 1, p = 0.046, v = 0.245$$χ2 = 3.967, df = 1, p = 0.046, v = 0.245$).

It also indirectly contributes to **Hypothesis 2** and **Hypothesis 3**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE with some or all N1s.

Actual outcome: the fact that the marginal significance disappears with 'Apfel' and 'Bucht' in the most "contaminated" dataset `ic_alt` indirectly points towards the effect of the problematic groups on the ExtPE. The fact that the (marginal) significance disappears with all three N1s in the moderate/high frequency settings indirectly points towards the effect of frequency on the ExtPE.

The significant results of N1s 'Apfel' and 'Person' confirm finding of Pankratz 2017. However, none of the other six "significant" N1s reported by Pankratz 2017 reach statistical significance in any of the nine data samples.

Study 3

Study 3 focuses on the distribution of non-pluralic and pluralic linker in compounds in the Sg form only. It contrasts the distribution in the "cleanest" `ic_alt_nipe_npll` vs the two other datasets with an increasing level of "contamination" by the problematic groups.

Study 3 contributes to **Hypothesis 2** and provides a more detailed account of the effect of the problematic groups on ExtPE. Expected outcome: more "contaminated" datasets will have a larger share of pluralic linkers in the Sg form (due to more possible sources).

We find that `ic_alt` has a larger share of pluralic linker in the compounds in the Sg form, reaching a statistical significance at sig = 0.005.

`chi2_summary`

	vs_ic_alt_nipe	vs_ic_alt
N	1607.000	1819.000
df	1.000	1.000
chi2	0.578	10.394
p	0.447	0.001
cramer_v	0.019	0.076
resids_cong	0.812	3.274
resids_incong	-0.812	-3.274

Study 3 Results

> Summary: Study 3 provides direct evidence **for Hypothesis 2**.

Study 3 contributes to **Hypothesis 2**. Expected outcome: more "contaminated" datasets will have a larger share of pluralic linkers in the Sg form (due to more possible sources).

Actual outcome: while comparison of the linker distributions over the compounds in the Sg form in `ic_alt_nipe_npll` vs `ic_alt_nipe` did not produce statistical significance, comparing the "cleanest" `ic_alt_nipe_npll` and the most "contaminated" `ic_alt` reveals a statistically significant effect at sig = 0.005 ($χ2 = 10.394, df = 1, p = 0.001, v = 0.076$$χ2 = 10.394, df = 1, p = 0.001, v = 0.076$). The examination of the symmetrical (standardised) residuals reveals that the diverging data points followed the expected direction, that is, compounds in the Sg form in `ic_alt` have a larger share of pluralic linkers, and those in `ic_alt_nipe_npll` a smaller share of pluralic linkers, than expected by chance ($3.274$$3.274$). This means that compounds in `ic_alt` have more sources for the pluralic linker to appear in the Sg form. The finding supports the indirect evidence for Hypothesis 2 from Studies 1 and 2, and confirms the effect.

Study 4

Study 4 focuses on the distribution of non-pluralic and pluralic linker in compounds in the Pl form only. It contrasts the distribution in the low frequency setting vs the moderate/high frequency setting, within each of the three dataset.

Study 4 contributes to **Hypothesis 3** and provides a more detailed account of the effect of the frequency setting on ExtPE. Expected outcome: the share of the pluralic linkers in compounds in the Pl form will be larger in in the low frequency setting (since most of the ExtPE manifestations are a product of immediate language).

```
# 3D res DF: dataset x freq pairing x chi2 test metrics
chi2_summary = pd.concat(chi2_summaries, keys=keys)  \
    .swaplevel(0, 1)    \
    .sort_index(level=0, sort_remaining=False)  \
    .reindex(chi2df.index, level=0)
savetsv(chi2_summary, st4dir, "chi2_summary")
chi2_summary
```

		vs_mfreq	vs_hfreq
N	ic_alt_nipe_npll	1010.000	953.000
	ic_alt_nipe	1580.000	1494.000
	ic_alt	1876.000	1772.000
df	ic_alt_nipe_npll	1.000	1.000
	ic_alt_nipe	1.000	1.000
	ic_alt	1.000	1.000
chi2	ic_alt_nipe_npll	4.376	7.755
	ic_alt_nipe	1.718	2.933
	ic_alt	0.939	1.952
p	ic_alt_nipe_npll	0.036	0.005
	ic_alt_nipe	0.190	0.087
	ic_alt	0.332	0.162
cramer_v	ic_alt_nipe_npll	0.066	0.090
	ic_alt_nipe	0.033	0.044
	ic_alt	0.022	0.033
resids_cong	ic_alt_nipe_npll	2.158	2.854
	ic_alt_nipe	1.363	1.767
	ic_alt	1.016	1.446
resids_incong	ic_alt_nipe_npll	-2.158	-2.854
	ic_alt_nipe	-1.363	-1.767
	ic_alt	-1.016	-1.446

The only comparison that produces statistically significant results at sig = 0.05 is that in `ic_alt_nipe_npll`.

```
chi2_summary.loc[["p", "cramer_v"]]
```

		vs_mfreq	vs_hfreq
p	ic_alt_nipe_npll	0.036	0.005
	ic_alt_nipe	0.190	0.087
	ic_alt	0.332	0.162
cramer_v	ic_alt_nipe_npll	0.066	0.090
	ic_alt_nipe	0.033	0.044
	ic_alt	0.022	0.033

Study 4 Results

> Summary: Study 4 provides direct evidence **for Hypothesis 3** and indirect evidence **for Hypothesis 2**.

Study 4 contributes to **Hypothesis 4**. Expected outcome: the share of the pluralic linkers in compounds in the Pl form will be larger in in the low frequency setting (since most of the ExtPE manifestations are a product of immediate language).

Actual outcome: comparison of the linker distributions over the compounds in the Pl form in difference frequency settings produced statistical significance only in the `ic_alt_nipe_npll` dataset: low vs. moderate frequency reaches significance at sig = 0.05, and low vs. high frequency at sig = 0.005 (low vs. moderate: $χ2 = 4.376, df = 1, p = 0.036, v = 0.066$$χ2 = 4.376, df = 1, p = 0.036, v = 0.066$; low vs. high: $χ2 = 7.755, df = 1, p = 0.005, v = 0.09$$χ2 = 7.755, df = 1, p = 0.005, v = 0.09$). The examination of the symmetrical (standardised) residuals reveals that the diverging data points followed the expected direction, that is, compounds in the Pl form inserted pluralic linkers more frequently in the low frequency setting and less frequently in the moderate/high than expected by chance ($2.158$$2.158$ and $2.854$$2.854$, respectively). This finding supports the indirect evidence for Hypothesis 3 from Studies 1 and 2, and confirms the effect.

The fact that the more "contaminated" datasets did not exhibit a significant difference additionally speaks for Hypothesis 2 and may be explained by that the additional sources of pluralic linkers there erode the effect. This is attested by the fact that, while insignificant, the p-values still decrease, and the Cramer's V values increase there consistently with increasing the frequency level.